FINAL PRESENTATION

# What Microsoft's job postings reveal

182 engineering postings · published by Microsoft under MIT · no posting dates


| | |
|---|---|
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| Repository | github.com/Nayab-khalid/cadetX-Competitor-Demand-Prediction |

THE HEADLINE

# Three findings

- **Microsoft hires cloud-first; AI is present but secondary.** Azure appears in 67.6% of postings, Machine Learning in 30.2%. Reliability, networking, security and storage all cluster inside Azure work - the cloud is where the other disciplines happen, not a line item beside them.
- **The direction of travel is systems and security.** Holding role family fixed, C++ and Cybersecurity both rose +24.1 points from the oldest to the newest cohort, Apache Spark +12.0; Agile fell -30.6.
- **Seniority buys coordination, not syntax.** Technical Program Management is in 6.2% of Mid postings and 29.2% of Principal ones.

### And one finding about the method

Microsoft has the richest description text in the team (96.7% full-text coverage) and is nearly invisible on the basis the team compares with (5.5% of titles). Reading it also exposed a defect in the shared taxonomy that falsely tags 80 of 182 Microsoft postings.

TASKS 1 & 2

# No legal route to scrape - so Microsoft's own published sample

| Source | Expected | Actual | Verdict |
|---|---|---|---|
| jobs.careers.microsoft.com | live postings with dates | automated request refused (robots-disallowed) | **Rejected** |
| LinkedIn postings dataset | 33,246 rows, cross-market | zero Microsoft-employer postings - only consultancies | **Rejected** |
| Wayback Machine snapshots | recover posting dates | proposed in Task 1, not yet attempted | Open |
| microsoft/SkillsExtractorCognitiveSearch | sample data for an Azure demo | 182 postings, full text, MIT, published by Microsoft | **Used** |

No scraping and no licence ambiguity. The cost: no posting dates, all 182 postings are Engineering, and 181 of 182 are in Washington, USA.

**A correction to my own Task 2 report**

The report said all records had been inspected and held no contact information. That was wrong: three descriptions contained **askstaff@microsoft.com**, which the team's data schema forbids outright. The re-issued file replaces them with a visible [EMAIL REDACTED] marker and follows the shared schema. The validator now passes on everything except posting_date, which this source does not have.

TASKS 3 & 4

# The richest text in the team - and the least visible

| | NVIDIA | Google | Meta | Microsoft |
|---|---|---|---|---|
| Skills per posting, titles | 1.03 | 0.36 | 0.17 | **0.08** |
| Coverage, titles | 71.7% | 30.3% | 16.3% | **5.5%** |
| Coverage, full text | no text | 73.6% | 88.5% | **96.7%** |

- **Boilerplate is 29.1% of the corpus** - EEO, benefits and security-screening text in roughly 180 of 182 postings. Left in, it dominates any similarity measure while saying nothing about hiring.
- **My first cleaning pass silently destroyed text.** An unanchored pattern ran to the end of the document: one 3,897-character posting collapsed to 14 tokens. Every pattern is now anchored, and a guard **raises** if any posting keeps under 45% of its characters. Every member has boilerplate, so every member is exposed to the same failure.
- **My first extraction run reported 0.0% title coverage** and I nearly wrote it up as a finding. The shared matcher is case-sensitive and returns zero, not an error, on Title Case input.

Full text after removing the 'drive' false positive: 5.95 skills per posting (6.38 as the taxonomy stands). The NVIDIA deck shows 6.84 for Microsoft; the figures here are the committed Task 4 and Task 6 outputs, to reconcile with Nayab.

TASK 4

# A defect in the shared taxonomy, affecting everyone

| Company | Postings | Falsely tagged 'Autonomous Vehicles' | Why |
|---|---|---|---|
| Microsoft | 182 | **80 (44.0%)** | full text, alias 'drive' |
| Meta | 209 | **70 (33.5%)** | full text, alias 'drive' |
| Google | 1,127 | 0% - latent, not absent | Task 6 matched Google on titles only |
| NVIDIA | 1,745 | n/a | no description text |

Every one of those postings contains the ordinary verb - *"drive features to closure"*, *"drive impact with a global team"*, *"Drive for results"*. None is about vehicles. I found it by reading 20 postings by hand, not from the code.

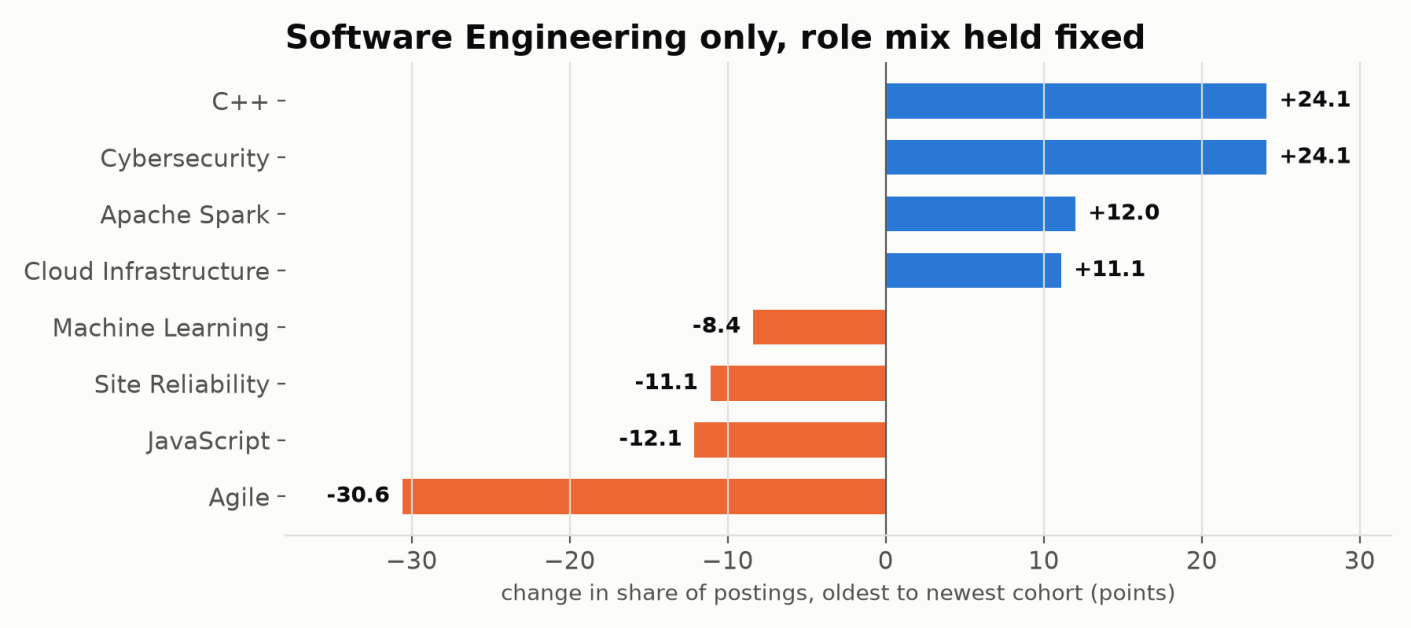
- Two more aliases were measured before proposing them and rejected: **teams** (96 hits, 4 are Microsoft Teams) and **windows** (60 hits, 11 are the operating system).
- Task 4 reports both numbers - as the taxonomy stands, so nothing already committed becomes incomparable, and with the alias suppressed, so the size of the error is visible.

**The fix is ready but not merged**

It is on its own branch, taxonomy/microsoft-extension, with Microsoft's proposed extension. It changes every member's Autonomous Vehicles numbers, so it needs the team's agreement before it goes to main.

TASK 5

# The skill that was rising was the role mix moving

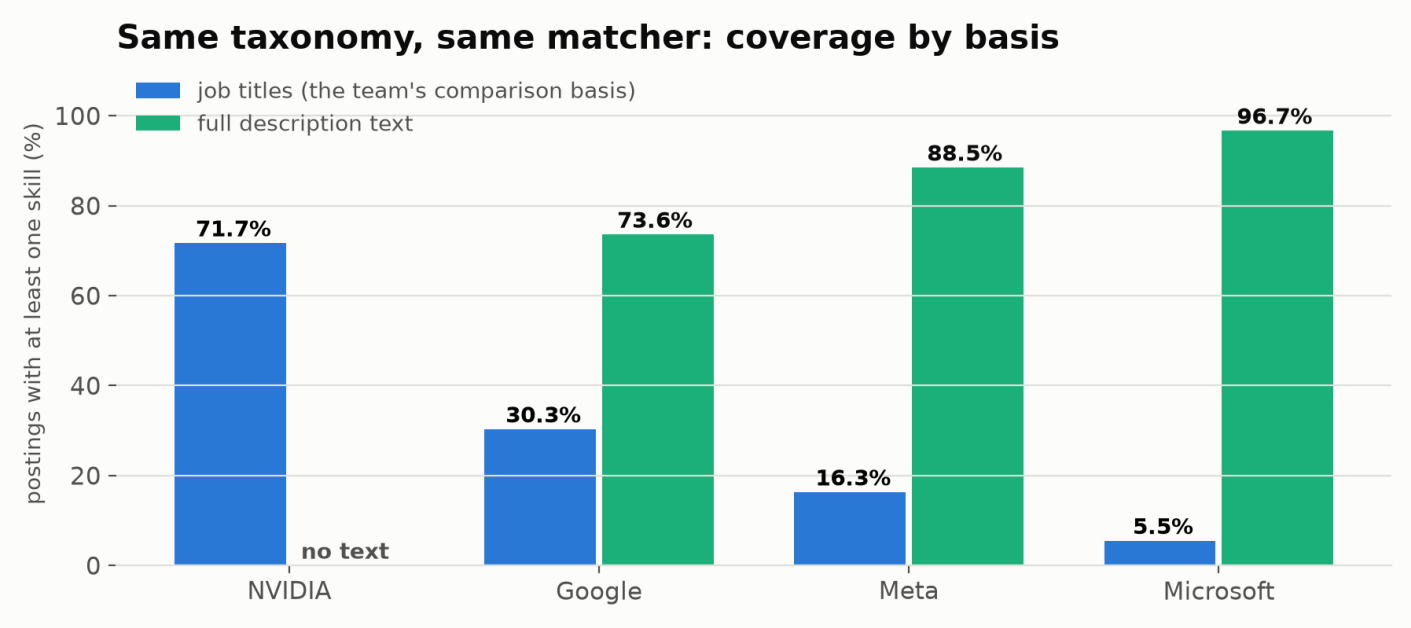



The first pass found Program Management skills up 17 points. The cohorts did not hold the same roles: Software Engineering fell from 78.3% to 58.7% of postings while Program Management rose from 2.2% to 17.4%. **Holding Software Engineering fixed, the Program Management effect disappears.** What survives is a coherent cluster: systems, security and data-platform work rising together.

No posting dates exist, so cohorts are cut on Microsoft's sequential requisition IDs: four equal groups by rank. This gives direction of change, not speed - the four cohorts could span four months or four years - and assumes IDs rise with time, which is standard but unverified. Agile's -30.6 may be job-ad style changing, not teams.

TASK 6

# On the team's comparison basis, Microsoft is nearly invisible




**An empty Microsoft column means naming, not hiring less**

Microsoft titles are generic by house style - "Software Engineer II", "Principal Data Scientist". NVIDIA's carry the technology - "Senior Linux Kernel Systems Software Engineer". Titles are still the right basis: they are the only one all four companies have. But every four-way chart needs this sentence beside it, or Microsoft reads as hiring narrower. Microsoft's taxonomy extension is deliberately not applied here: doing it alone would inflate Microsoft for the same reason NVIDIA is inflated today.

TASK 7

# A forecast on a calendar that does not exist

| Method | Verdict | Reason |
|---|---|---|
| Moving average | **Chosen** | 24 monthly points: stable, transparent, hard to overfit |
| ARIMA(1,0,0) | Cross-check | wants 30-50 points for its parameter search to mean much |
| Prophet | Rejected | built for multi-year daily series; on 24 points it fits noise |
| Deep learning | Rejected | 182 postings would be memorised |

| Month (simulated) | Moving average | ARIMA(1,0,0) | 1-sd band |
|---|---|---|---|
| 2026-01 | 7.3 | 6.7 | 3.3 - 11.4 |
| 2026-02 | 8.1 | 7.7 | 4.1 - 12.1 |
| 2026-03 | 8.8 | 7.5 | 4.8 - 12.9 |

**Every number on this slide describes the simulation**

Dates were seeded in requisition-ID order, following Google's Task 3 precedent: the order is real, the spacing is invented. Filenames carry SIMULATED and the output has a basis column saying so. The two methods agree, which proves the implementation, not the forecast: the residual spread (4.04 on a mean of 7.6) is more than half the level.

TASK 8

# Three measures, three different answers

| Pair | Jaccard (same skills?) | Cosine, skill shares | Cosine, title text |
|---|---|---|---|
| Microsoft - NVIDIA | 0.083 | 0.324 | 0.622 |
| NVIDIA - Meta | 0.203 | 0.173 | 0.218 |
| Microsoft - Meta | 0.059 | 0.170 | 0.092 |
| NVIDIA - Google | 0.400 | 0.116 | 0.196 |
| Microsoft - Google | 0.192 | 0.092 | 0.118 |
| Google - Meta | 0.321 | 0.029 | 0.072 |

- **NVIDIA - Google:** highest Jaccard, near-lowest cosine. They name the same skills and weight them completely differently. One measure alone would have called them the closest pair.
- **Microsoft - NVIDIA:** the opposite shape. Few shared skill names, but the closest proportions and the most alike title language. Microsoft's few title hits fall where NVIDIA concentrates.
- **Two sparse companies can look similar by both being sparse.** With 5.5% title coverage, a handful of postings moves Microsoft's vectors. I agree with the NVIDIA deck: do not publish a four-company similarity matrix yet.

TASK 9

# What a decision-maker should do with this

| If you are | Then |
|---|---|
| Competing for cloud engineers | Hire for reliability, security and networking inside Azure, not cloud on its own. Azure appears in 32 of 34 Site Reliability postings, 29 of 31 Networking, 50 of 56 Cybersecurity. |
| Watching Microsoft's direction | Systems, security and data-platform skills rising together (C++, Cybersecurity, Spark); Agile and JavaScript falling. Order only - not when, not how fast. |
| Hiring senior people | Principal roles ask for programme management, not more languages: TPM 29.2% at Principal. Only 24 Principal postings - each moves it by ~4 points. |
| Sizing Microsoft's AI hiring | In engineering postings AI is secondary: Machine Learning 30.2% against Azure 67.6%. |
| Benchmarking salary or region | Not possible. No salary field; 181 of 182 postings are in Washington, USA. |

HONESTY

# What this data cannot tell you

- **When, or how fast.** The source has no posting dates. Velocity and seasonality are not measured; the Task 7 forecast is on a synthetic calendar and labelled as such everywhere it appears.
- **Market share.** 182 postings Microsoft published as demo data for an Azure project - a sample with an unknown selection, not a census of hiring.
- **Company-wide strategy.** All 182 are Engineering. Sales, Finance, Marketing and Operations are absent.
- **Regional patterns.** 181 of 182 are in Washington, USA.
- **Capability on the titles basis.** A skill missing from a Microsoft title means the title did not name it - Microsoft titles rarely name anything.
- **The cohort ordering, if requisition IDs are not chronological.** Every Task 5 finding rests on that assumption.

### Why this slide exists

Every limitation here is in the committed task notes. Every number in this deck comes from a committed output file - read directly by build_deck.py - or from the Task 2-9 notes. A mentor can check any figure against the repository.

NEXT

# What the team should do, and questions

| Priority | Action | Unblocks |
|---|---|---|
| 1 | Agree the 'drive' alias fix; every member extends the taxonomy for their own company; all re-run Task 4 | Tasks 6 and 8; removes a 44% false tag from Microsoft |
| 2 | Decide titles or full text once, for everyone | Like-for-like comparison; Microsoft visible |
| 3 | Add the over-deletion guard to every member's Task 3 | Silent text loss surfacing later in Task 8 |
| 4 | Recover Microsoft dates from Wayback Machine snapshots | Real time axis for Tasks 5 and 7 |
| 5 | Run Task 12 on hand-corrected labels | Skills the alias list cannot see |

Expected questions: why a 182-posting demo sample · whether requisition IDs are chronological · why Agile fell · why the Microsoft taxonomy extension was not applied · how the Task 2 report missed the email addresses.